CSE 2022

1. a) Use Taylor's polynomial to derive the orders of error for the Forward Difference rule, Backward Difference rule, and Central Difference rule of Numerical Differentiation. Show mathematical reasoning for choosing one of these three. 9+5 (CO1, CO3)

Errors of approximation

From taylor series,

$$f(x+h) = f(x) + f'(x)\cdot h + \frac{f''(c)}{2!}h^2, \quad x < c < x+h$$

For forward difference, ——— ①

$$\Rightarrow f(x_{i+1}) = f(x+h) = f(x_i) + f'(x_i)h + \frac{f''(c)}{2!}h^2$$

$$\Rightarrow f'(x_i) = \frac{f(x_{i+1}) - f(x_i) - \frac{f''(c)}{2!}h^2}{h}$$

$$= \frac{f(x_{i+1}) - f(x_i)}{h} - \frac{f''(c)}{2!}h$$

But, for forward difference, $f'(x_i) = \frac{f(x_{i+1}) - f(x_i)}{h}$

Thus, error $\rightarrow \frac{-f''(c)}{2!}h \rightarrow O(h)$

For backward difference,

$$f(x_{i-1}) = f(x-h) = f(x_i) - f'(x_i)h + \frac{f''(c)}{2!}h^2$$

$$\Rightarrow f'(x_i) = \frac{f(x_i) - f(x_{i-1})}{h} - \frac{f''(c)}{2!}h$$

But backward difference, $f'(x_i) = \frac{f(x_i) - f(x_{i-1})}{h}$

Thus error $\rightarrow \frac{-f''(c)}{2!}h \rightarrow O(h)$

For central difference, Using Taylor 3rd degree

$$f(x_{i+1}) = f(x+h) = f(x_i) + f'(x_i)h + \frac{f''(x_i)}{2!}h^2 + \frac{f'''(c_1)}{3!}h^3 \quad \text{—①}$$

and $$f(x_{i-1}) = f(x-h) = f(x_i) - f'(x_i)h + \frac{f''(x_i)}{2!}h^2 - \frac{f'''(c_2)}{3!}h^3 \quad \text{—②}$$

① − ② ⇒

$$f(x_{i+1}) - f(x_{i-1}) = 2f'(x_i)h + \frac{f'''(c_1)}{3!}h^3 + \frac{f'''(c_2)}{3!}h^3$$

$$\Rightarrow f'(x_i) = \frac{f(x_{i+1}) - f(x_{i-1})}{2h} - \frac{f'''(c_1) + f'''(c_2)}{3!} \times \frac{h^2}{2}$$

But central difference formula is $f'(x_i) = \frac{f(x_{i+1}) - f(x_{i-1})}{2h}$

Thus error → $\frac{f'''(c_1) + f'''(c_2)}{3!} \times \frac{h^2}{2}$ → $O(h^2)$

Reasoning:

Here, Error in central difference, $O(h^2)$ is much lower than $O(h^2)$ of forward or backward difference.

b) A tissue paper company advertises that every roll of their toilet paper has at least 250 sheets. The probability that there are 250 or more sheets in the toilet paper roll is given by the following equation: 11 (CO2) (PO2)

$$P(y \geq 250) = \int_{250}^{\infty} 0.3515e^{-0.3881(y-252.2)^2} dy$$

By theoretical reasoning, the above integral calculates the probability up to infinite sheets. But practically, it has been seen that the number of sheets rarely exceed 270. So, considering 270 as the maximum possible number of sheets, the above integral can be approximated as follows:

$$P(y \geq 250) = \int_{250}^{270} 0.3515e^{-0.3881(y-252.2)^2} dy$$

Use the 2-segment Trapezoidal rule to find the probability that there are 250 or more sheets in a random toilet paper roll.

For multi segment trapezoidal rule,

$$I = \frac{b-a}{2n}\left[f(a) + 2\left\{ \sum_{i=1}^{n-1} f(a+ih) \right\} + f(b) \right] ; \quad n \rightarrow \text{seg no}$$

Now,

$$h = \frac{b-a}{2}$$

$$= \frac{270-250}{2}$$

$$= 10$$

$$f(x) = 0.3515e^{-0.3881(y-252.2)^2}$$

$$\therefore f(250) = 0.0537211$$

$$\therefore f(260) = 1.955970\,4 \times 10^{-11}$$

$$\therefore f(270) = 1.388807484 \times 10^{-54}$$

$$I = \frac{270-250}{2\times 2}\left[f(250) + 2f(250+10) + f(270) \right]$$

$$= \frac{20}{4}\left[0.0537211 \right]$$

$$= 0.2686$$

2. a) Without using linearization transform, describe the individual solution approaches for the following nonlinear regression models: 4+5+4 (CO1) (PO1)
   i. Exponential model
   ii. Growth model
   iii. Polynomial model

i) Exponential Model:

$y = ae^{bx}$ thus $E_i = y_i - ae^{bx_i}$

$\therefore S_r = \sum_{i=1}^{n} (y_i - ae^{bx_i})^2$

Now, $\frac{\partial S_r}{\partial a} = \sum_{i=1}^{n} 2(y_i - ae^{bx_i}).(-e^{bx_i})$ ——— ①

And $\frac{\partial S_r}{\partial b} = \sum_{i=1}^{n} 2(y_i - ae^{bx_i})(-ax_i e^{bx_i})$ ——— ②

① → $\frac{\partial S_r}{\partial a} = 0$

$\Rightarrow \sum_{i=1}^{n} -e^{bx_i} y_i + a \sum_{i=1}^{n} e^{2bx_i} = 0$

$\Rightarrow a = \frac{\sum_{i=1}^{n} y_i e^{bx_i}}{\sum_{i=1}^{n} e^{2bx_i}}$ ——— ③

Again, ② →

$\frac{\partial S_r}{\partial b} = 0$

$\Rightarrow -\sum_{i=1}^{n} y_i a x_i e^{bx_i} + a^2 \sum_{i=1}^{n} x_i e^{2bx_i} = 0$

$\Rightarrow \sum_{i=1}^{n} y_i x_i e^{bx_i} - a \sum_{i=1}^{n} x_i e^{bx_i} = 0$

$\Rightarrow \sum_{i=1}^{n} y_i x_i e^{bx_i} - \frac{\sum_{i=1}^{n} y_i e^{bx}}{\sum_{i=1}^{n} e^{2bx_i}} \sum_{i=1}^{n} x_i e^{bx_i} = 0$ → Solve via bisection or secant

ii) Growth Model

$$y = \frac{1}{1+be^{-cx_i}} \quad \text{thus,} \quad E_i = y_i - \frac{1}{1+be^{-cx_i}} \qquad S_r = \Sigma E_i^2$$

find $\frac{\partial S_r}{\partial a} = \frac{\partial S_r}{\partial b} = \frac{\partial S_r}{\partial c} = 0$,

put a, b, c, values. Then solve with Newton Raphson.

iii) Polynomial Model

$$y = a_0 + a_1 x + a_2 x^2 + \ldots + a_m x^m$$

$$E = y - (a_0 + a_1 x + a_2 x^2 + \ldots + a_m x^m)$$

$$S_r = \Sigma E^2$$

$$\frac{\partial S_r}{\partial a_0} = \frac{\partial S}{\partial a_1} = \frac{\partial S}{\partial a_2} = \ldots = \frac{\partial S}{\partial a_m} = 0$$

Solve for $a_0, a_1, \ldots, a_m$

Then use Newton Raphson

b) What is the drawback in directly using the first 3 terms of the Taylor series for deriving y_{i+1} in Runge-Kutta 2nd order method? What is the alternative way of approximating the first 3 terms of the series? 3+3 (CO3) (PO2)

The main drawback is that, calculating the 2nd derivative is very expensive

To solve this, Runge-Kutta used the following method:

$$y_{i+1} = y_i + (a_1 k_1 + a_2 k_2) h$$

Where, $k_1 = f(x, y)$

$$k_2 = f(x + p_1 h, y + q_1 k_1 h)$$

This form allows us to take advantage of 2nd order without calculating $f'(x, y)$

Here, $a_1 + a_2 = 1$

$a_1 p_1 = \frac{1}{2}$

$a_1 q_1 = \frac{1}{2}$

$a_2 = \frac{1}{2}$ Heun, $\frac{2}{3}$ Ralston, 1 Midpoint } Method

c) What are the different forms of the approximation of Runge-Kutta 2nd order method based on Heun's method, Midpoint method and Ralston's method? 6 (CO1) (PO1)

Runge-Kutta used the following for 2nd order:

$$y_{i+1} = y_i + (a_1 k_1 + a_2 k_2) h$$

Where, $k_1 = f(x, y)$

$$k_2 = f(x + p_1 h,\ y + q_1 k_1 h)$$

Here, $a_1 + a_2 = 1$

$a_1 p_1 = \frac{1}{2}$

$a_1 q_1 = \frac{1}{2}$

$a_2 = \begin{cases} \frac{1}{2} & \text{Heun} \\ \frac{2}{3} & \text{Ralston} \\ 1 & \text{Midpoint} \end{cases}$ Method

Then put a_2 value in the eqn the get values of k. finally get values for y_i

3. a) Find the Eigenvalues and Eigenvectors of the following matrix A: 12 (CO2) (PO1)

$$A = \begin{bmatrix} 2 & 27 & 0 \\ 0 & 4 & 40 \\ 0 & 3 & 30 \end{bmatrix}$$

$Av = \lambda v$

$\Rightarrow |A - I\lambda| = 0$

$$\Rightarrow \begin{vmatrix} 2-\lambda & 27 & 0 \\ 0 & 4-\lambda & 40 \\ 0 & 3 & 30-\lambda \end{vmatrix} = 0$$

$\Rightarrow (2-\lambda)\{(4-\lambda)(30-\lambda) - 120\} - 27\{0 - 0\} + 0\{0 - 0\} = 0$

$\Rightarrow (2-\lambda)\{120 - 4\lambda - 30\lambda + \lambda^2 - 120\} = 0$

$\Rightarrow (2-\lambda)\{\lambda^2 - 34\lambda\} = 0$

$\Rightarrow \lambda(\lambda-2)(\lambda-34) = 0$

$\therefore \lambda = 0, 2, 34$

Now, $(A - I\lambda)V = 0$

For $\lambda = 0$,

$$\left[\begin{array}{ccc|c} 2 & 27 & 0 & 0 \\ 0 & 4 & 40 & 0 \\ 0 & 3 & 30 & 0 \end{array}\right]$$

$$\Rightarrow \left[\begin{array}{ccc|c} 1 & 13.5 & 0 & 0 \\ 0 & 1 & 10 & 0 \\ 0 & 1 & 10 & 0 \end{array}\right]$$

$$\Rightarrow \left[\begin{array}{ccc|c} 1 & 13.5 & 0 & 0 \\ 0 & 1 & 10 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$$= \left[\begin{array}{ccc|c} 1 & 0 & 135 & 0 \\ 0 & 1 & 10 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$$\Rightarrow \begin{matrix} x_1 + 135x_3 = 0 \\ x_2 + 10x_3 = 0 \end{matrix} \Big\} \begin{matrix} x_1 = -135x_3 \\ x_2 = -10x_3 \end{matrix}$$

Thus, $v = \begin{bmatrix} -135 \\ -10 \\ 1 \end{bmatrix} k$

For, $\lambda = 2$,

$$\left[\begin{array}{ccc|c} 0 & 27 & 0 & 0 \\ 0 & 2 & 40 & 0 \\ 0 & 3 & 28 & 0 \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} 0 & 1 & 0 & 0 \\ 0 & 1 & 20 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right]$$

$$\rightarrow \left[\begin{array}{ccc|c} 0 & 1 & 0 & 0 \\ 0 & 1 & 20 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

So, $x_2 = 0$

$x_2 + 20x_3 = 0 \Rightarrow \begin{matrix} x_2 = 0 \\ x_3 = 0 \end{matrix} \Rightarrow v = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} k$

For $\lambda = 34$

$$\left[\begin{array}{ccc|c} -32 & 27 & 0 & 0 \\ 0 & -30 & 40 & 0 \\ 0 & 3 & -4 & 0 \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & -\frac{27}{32} & 0 & 0 \\ 0 & 1 & -\frac{4}{3} & 0 \\ 0 & -\frac{3}{4} & 1 & 0 \end{array}\right]$$

$$\to \left[\begin{array}{ccc|c} 1 & 0 & -\frac{9}{8} & 0 \\ 0 & 1 & -\frac{4}{3} & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \to \begin{array}{l} x_1 - \frac{9}{8}x_3 = 0 \\ x_2 - \frac{4}{3}x_3 = 0 \end{array} \Bigg\} v = \begin{bmatrix} 9/8 \\ 4/3 \\ 1 \end{bmatrix} k$$

b) Describe how linearization transform is applied on the following nonlinear models: 2+2+2

i. Exponential model (CO1)

ii. Power model (PO1)

iii. Growth model

i) Exponential

$$y_i = a e^{bx_i}$$

$$\Rightarrow \ln y_i = \ln\left(a e^{bx_i}\right)$$

$$= \ln a + \ln e^{bx_i}$$

$$= \ln a + bx_i$$

$$\therefore \quad z = a_0 + a_1 x_i$$

$$a_1 = \frac{n\Sigma x_i y_i - \Sigma x \Sigma y}{n\Sigma x^2 - (\Sigma x)^2}$$

$$a_0 = \bar{y} - a_1 \bar{x}$$

ii) Power model

$$y = ax^b$$

$$\ln y = \ln a + b \ln x$$

$$\Rightarrow z = a_0 + a_1 x_i$$

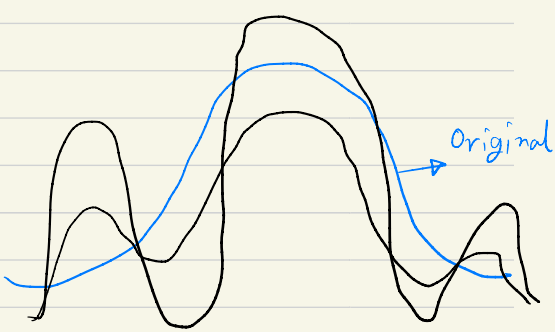
c) Why do we normally use polynomials as interpolating functions? Can single polynomials always be used to accurately interpolate data? 2+5 (CO3) (PO2)

An n-degree polynomial can pass through $n+1$ points. It is a common choice for interpolation because

1) Easy to evaluate
2) " " differenciate
3) " " integrate

However single polynomials are not always suitable due to oscillation

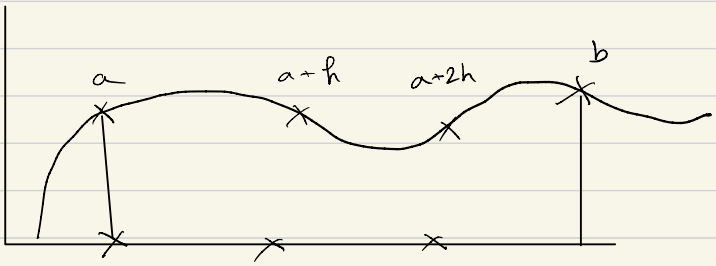
$$y = \frac{1}{1+25x^2}$$



So, lots of lower order polynomials are better suited → Spline

4. a) For approximating $I = \int_a^b f(x)dx$ using Simpson's 3/8 rule, how are the points chosen over the interval $[a, b]$, and what are their exact coordinates in terms of a and b? Provide necessary illustrations. 9 (CO1) (PO1)

We use a 3rd order polynomial



$h = \frac{b-a}{3}$;

points $(a, f(a))$, $(a+h, f(a+h))$, $(a+2h, f(a+2h))$, $(b, f(b))$

b) The minimum number of segments for Simpson's 3/8 rule is 3. Given that the 3-segment formula is $I = \frac{3h}{8}[f(x_0) + 3f(x_1) + 3f(x_2) + f(x_3)]$, derive the general n-segment formula? 8 (CO1) (PO2)

$h = \frac{b-a}{n}$; n = # of segments

$$I = \int_a^b f_3(x)\,dx$$

$$= \int_a^b f_3(x)\,dx$$

$$= \int_{x_0=a}^{x_3} f_3(x)\,dx + \int_{x_3}^{x_6} f_3(x)\,dx + \cdots + \int_{x_{n-3}}^{x_n = b} f_3(x)\,dx$$

$$= \frac{3h}{8}\left[f(x_0) + 3f(x_1) + 3f(x_2) + f(x_3)\right]$$
$$+ \frac{3h}{8}\left[f(x_3) + 3f(x_4) + 3f(x_5) + f(x_6)\right]$$
$$+ \cdots$$
$$+ \frac{3h}{8}\left[f(x_{n-3}) + 3f(x_{n-2}) + 3f(x_{n-1}) + f(x_n)\right]$$

$$= \frac{3h}{8}\left[f(x_0) + 3\sum_{i=1,4,7}^{n-2} f(x_i) + 3\sum_{i=2,5,8}^{n-1} f(x_i) + 2\sum_{i=3,6}^{n-3} f(x_i) + f(x_n)\right]$$

c) What is the LU decomposition of A, where $A = \begin{bmatrix} 1 & 1 & -1 \\ 1 & -2 & 3 \\ 2 & 3 & 1 \end{bmatrix}$ — 8 (CO2) (PO1)

$$\begin{bmatrix} 1 & 1 & -1 \\ 1 & -2 & 3 \\ 2 & 3 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

$$= \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ l_{21}u_{11} & l_{21}u_{12}+u_{22} & l_{21}u_{13}+u_{23} \\ l_{31}u_{11} & l_{31}u_{12}+l_{32}u_{22} & l_{31}u_{13}+l_{32}u_{23}+u_{33} \end{bmatrix}$$

Here, $u_{11} = 1$, $u_{12} = 1$, $u_{13} = -1$

$l_{21}u_{11} = 1 \Rightarrow l_{21} = 1$

$l_{21}u_{12} + u_{22} = -2 \Rightarrow u_{22} = -3$

$l_{21}u_{13} + u_{23} = 3 \Rightarrow u_{23} = 4$

$l_{31}u_{11} = 2 \Rightarrow l_{31} = 2$

$l_{31}u_{12} + l_{32}u_{22} = 3 \Rightarrow l_{32} = \frac{-1}{3}$

$2 \times 1 + -3\,l_{32} = 3$

$l_{31}u_{13} + l_{32}u_{23} + u_{33} = 1$

$\Rightarrow 2 \times (-1) + 4\left(-\frac{1}{3}\right) + u_{33} = 1$

$\therefore u_{33} = 13/3$

$\therefore LU =$

$$\begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 2 & -\frac{1}{3} & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & -1 \\ 0 & -3 & 4 \\ 0 & 0 & 13/3 \end{bmatrix}$$

5. a) What is the motivation behind the use of LU decomposition? Provide a scenario where the use of LU decomposition can enhance the performance of a numerical analysis system. 5+5 (CO3) (PO2)

Motivation:

For $Ax = b$; if we need to solve x for multiple b; performing Gaussian elimination each time can be very expensive.

But if we decompose it to $LUx = b$; we can easily calculate using

$Ly = b$ (Forward substitution) and

$Ux = y$ (Backward substitution)

Here, we dont need to perform gaussian elimination even when b changes.

Lets say we have $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$

and $b_1 = [1\ 2\ 3]^T$, $b_2 = [3\ 5\ 7]^T$, $b_3 = [6\ 1\ 3]^T$

Traditional way:

$Ax = b_1$

$$\left[\begin{array}{ccc|c} a_{11} & a_{12} & a_{13} & 1 \\ a_{21} & a_{22} & a_{23} & 2 \\ a_{31} & a_{32} & a_{33} & 3 \end{array}\right] \quad \left[\begin{array}{ccc|c} a_{11} & a_{12} & a_{13} & 3 \\ a_{21} & a_{22} & a_{23} & 5 \\ a_{31} & a_{32} & a_{33} & 7 \end{array}\right] \quad \left[\begin{array}{ccc|c} a_{11} & a_{12} & a_{13} & 6 \\ a_{21} & a_{22} & a_{23} & 1 \\ a_{31} & a_{32} & a_{33} & 3 \end{array}\right]$$

we need to perform Gaussian elemination for all values of b.

LU method:

$L(Ux) = b_1$ — easy & fast

$Ly = b_1 \rightarrow y$ (Lower triangular)

Now, $Ux = y \rightarrow x$ (Upper triangular) — easy & fast

Scenario:

Scenario: Solving Linear Systems in Real-Time Traffic Optimization

Consider a city's traffic management system which uses a complex network of sensors and real-time data feeds to optimize traffic flow across the city. The system needs to solve a large system of linear equations every few minutes to update the traffic light timings, estimate congestion, and suggest optimal routes to vehicles.

The system of equations that needs to be solved is based on the current state of the city's roads (congestion levels, travel times, etc.), which doesn't change dramatically from one minute to the next. As a result, the coefficient matrix A of the system Ax = b is almost the same for every set of equations that the system needs to solve.

Here, LU decomposition can greatly enhance the performance of the system. Instead of solving the system of equations from scratch every time, the system can perform an LU decomposition of the matrix A once and then use the result to solve the system for different b vectors.

Computing the LU decomposition has a time complexity of O(n^3), where n is the number of roads in the city. However, once the decomposition is computed, the system can solve the system of equations for any given vector b in O(n^2) time. Therefore, if the system needs to solve the equation for many different b vectors, using LU decomposition can be significantly more efficient than solving the system from scratch each time.

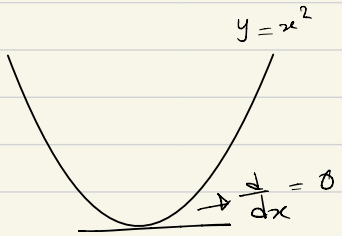
Moreover, if small changes occur in the city's traffic network, such as a road closure, the system can leverage techniques like rank-1 updates to adjust the LU decomposition, avoiding the need to recompute it entirely. This approach offers substantial computational savings and allows the traffic management system to operate in real-time, even as the city's traffic conditions change.

b) Given n data pairs $(x_1, y_1), \ldots, (x_n, y_n)$, do the values of the constants a_0 and a_1 in the least square regression line model $y = a_0 + a_1 x$ correspond to the absolute minimum of the sum of the squares of residuals? Are the constants of regression unique? 9+6 (CO1) (PO2)

For MSE, $y_i = a_0 + a_1 x_i$

$E = y_i - (a_0 + a_1 x_i)$

$S_r = \sum_{i=1}^{n} E^2$



Here, $\frac{\partial S_r}{\partial a_0} = 0$

and $\frac{\partial S_r}{\partial a_1} = 0$

} Solving

$$a_1 = \frac{n \Sigma x_i y_i - \Sigma x_i \Sigma y_i}{n \Sigma x_i^2 - (\Sigma x_i)^2}$$

$$a_0 = \frac{\Sigma x_i^2 \Sigma y_i - \Sigma x_i \Sigma x_i y_i}{n \Sigma x_i^2 - (\Sigma x_i)^2}$$

Here, a_0 and a_1 will be min when

$$\frac{\partial^2 S_r}{\partial a_0^2} \times \frac{\partial^2 S_r}{\partial a_0^2} - \left(\frac{\partial^2 S_r}{\partial a_0 \partial a_1} \right)^2 > 0 \quad \text{and} \quad \frac{\partial^2 S_r}{\partial a_0^2} > 0 \quad \text{or} \quad \frac{\partial^2 S_r}{\partial a_1^2} > 0$$

Now, $\frac{\partial S_r}{\partial a_0} = 2\Sigma\left(y_i - (a_0 + a_1 x_i)\right)(-1)$

$$\frac{\partial^2 S_r}{\partial a_0^2} = -2\Sigma(-1) = 2n$$

Again, $\frac{\partial S_r}{\partial a_1} = 2\Sigma\left(y_i - (a_0 + a_1 x_i)\right)(-x_i)$

$$\frac{\partial^2 S_r}{\partial a_1^2} = 2\Sigma x_i^2$$

Again, $\frac{\partial^2 S_r}{\partial a_0 \partial a_1} = -2\Sigma(-x_i) = 2\Sigma x_i$

Now, $\frac{\partial^2 S_r}{\partial a_0^2} \times \frac{\partial^2 S_r}{\partial a_1^2} - \left(\frac{\partial^2 S_r}{\partial a_0 \partial a_1}\right)^2$

$= 2n \times 2\Sigma x_i^2 - 4\left(\Sigma x_i\right)^2$

$= 4n\,\Sigma x_i^2 - 4\left(\Sigma x_i\right)^2$

$= 4\left(n\Sigma x_i^2 - \left(\Sigma x_i\right)^2\right)$

$= 4 \sum_{\substack{i=1 \\ i<j}}^{n} \left(x_i - x_j\right)^2 > 0$

Here, Local minima = Absolute minima

Since first derivative = 0 only at one point, the constants are unique

6. a) To maximize the catch of bass in a lake, it is suggested to throw the line to the depth of the thermocline. You can infer that there is a thermocline at a particular depth by noticing the sudden change in temperature. The following table provides us the depth vs temperature data for the lake: 6+7+3 (CO2, CO3) (PO3)

Table 1: Depth vs Temperature

Depth, h(m)	***Temperature, T(°C)***
0	19.1
-1	19.1
-2	19
-3	18.8
-4	18.7
-5	18.3
-6	18.2
-7	17.6
-8	11.7
-9	9.9
-10	9.1

It is noticed that there is a large difference in temperature from $h = -7$ to $h = -8$, so the thermocline must be somewhere between these two depths. The intermediate temperature data between these two depths is missing.

Determine the temperature at $h = -7.5$ using a **first order Lagrange polynomial** and then a **second order Lagrange polynomial** and find the absolute relative approximate error.

$$f_1(t) = \sum_{i=0}^{n} L_i \, f(t_i)$$

For 1st order, we choose 2 points (-7, 17.6) and (-8, 11.7)

Now,

$$L_0 = \prod_{\substack{j=1 \\ j \neq i}}^{1} \frac{t - t_j}{t_i - t_j} = \frac{t - (-8)}{(-7) - (-8)}$$

$$L_1 = \prod_{\substack{j=0 \\ j \neq i}}^{1} \frac{t - t_j}{t_i - t_j} = \frac{t - (-7)}{(-8) - (-7)}$$

Now, $f_1(t) = \frac{t+8}{-7+8} \times 17.6 + \frac{t+7}{-8+7} \times 11.7$

$\therefore f_1(-7.5) = 14.65^\circ C$

For 2nd order, we take 3 points $(-7, 17.6), (-8, 11.6), (-9, 9.9)$

Now, $f_2(t) = \sum_{i=0}^{n} L_i f(t)$

$$L_0 = \prod_{\substack{j=1 \\ j \neq 0}}^{2} \frac{x - x_j}{x_i - x_j} = \frac{x - (-8)}{-7 - (-8)} \times \frac{x - (-9)}{-7 - (-9)}$$

$$L_1 = \prod_{\substack{j=0 \\ j \neq 1}}^{2} \frac{x - x_j}{x_i - x_j} = \frac{x - (-7)}{-8 - (-7)} \times \frac{x - (-9)}{-8 - (-9)}$$

$$L_2 = \prod_{\substack{j=0 \\ j \neq 2}}^{2} \frac{x - x_j}{x_i - x_j} = \frac{x - (-7)}{-9 - (-7)} \times \frac{x - (-8)}{-9 - (-8)}$$

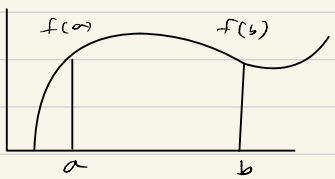
$f_2(t) = L_0 \times 17.6 + L_1 \times 11.6 + L_2 \times 9.9$

$= 14.0625$

b) Derive Simpson's 1/3 rule using the method of coefficients. Using necessary illustrations, determine the geometric interpretation of this derivation? 5+4 (CO1) (PO1)

For 1/3 rule, we use 2nd order polynomial

$f_2(x) = a_0 + a_1 x + a_2 x^2$



Now, Assume,

$$\int_a^b f(x)\,dx = c_1 f(a) + c_2 f\left(\frac{a+b}{2}\right) + c_3 f(b) \quad \text{—— (1)}$$

Now, the let right hand side be an exact expression for the integrals $\int_a^b 1\,dx$, $\int_a^b x\,dx$, $\int_a^b x^2\,dx$

So, $\int_a^b 1\,dx = b-a$

$$\Rightarrow b-a = c_1 + c_2 + c_3 \quad \text{—— (2)}$$

Again, $\int_a^b x = \frac{x^2}{2} = \frac{b^2-a^2}{2}$

$$\Rightarrow \frac{b^2-a^2}{2} = c_1 a + c_2 \frac{a+b}{2} + c_3 b \quad \text{— (3)}$$

Again, $\int_a^b x^2 = \frac{x^3}{3} = \frac{b^3 - a^3}{3}$

$$\Rightarrow \frac{b^3 - a^3}{3} = c_1 a^2 + c_2 \left(\frac{a+b}{2}\right)^2 + c_3 b^2 \quad \text{— (4)}$$

Solving (2), (3), (4) for c_0, c_1 and c_2 gives:

$$c_1 = \frac{b-a}{6}, \quad c_2 = \frac{2(b-a)}{3}, \quad c_3 = \frac{b-a}{6}$$

Now (1) →

$$\int_a^b f(x) = c_1 f(a) + c_2 f\left(\frac{a+b}{2}\right) + c_3 f(b)$$

$$= \frac{b-a}{6} f(a) + \frac{2(b-a)}{3} f\left(\frac{a+b}{2}\right) + \frac{b-a}{6} f(b)$$

$$= \frac{h}{3}\left[f(a) + 4 f\left(\frac{a+b}{2}\right) + f(b) \right]$$

SWE 2022

1. a) A Trunnion of diameter 12.363" has to be cooled from a room temperature of 80°F before it can be shrink-fit into a steel hub. The equation that gives the diametric contraction (in inches) of the trunnion in dry-ice (boiling point −108°F) is: 8+10 (CO2) (PO1)

$$\Delta D = 12.363 \int_{80}^{-108} (-1.2278 \times 10^{-11} T^2 + 6.194 \times 10^{-9} T + 6.015 \times 10^{-6}) dT$$

Here, a positive ΔD indicates expansion and negative value indicates contraction.
Find ΔD using the 2-segment and 4-segment Simpson's 1/3 rule.

$f(T) = (-1.2278 \times 10^{-11} T^2 + 6.194 \times 10^{-9} T + 6.015 \times 10^{-6})$

Using 2 segment,

$h = \frac{b-a}{n} = \frac{-108-80}{2} = -94$

$$I = \frac{h}{3}\left[f(a) + 4\sum_{i=1}^{n-1} f(a+ih) + f(b) \right]$$

$$= \frac{-94}{3}\left[f(80) + 4(80-94) + f(-108) \right]$$

$f(80) = 6.43194 \times 10^{-6}$

$f(-14) = 5.92588 \times 10^{-6}$

$f(-108) = 5.20284 \times 10^{-6}$

$\therefore I = -1.10727 \times 10^{-3}$

$\therefore \Delta D = -0.0137$

Using 4 segments,

$$h = \frac{b-a}{n} = \frac{-108-80}{4} = -47$$

$$I = \frac{h}{3}\left[f(a) + 4\sum_{\substack{i=1 \\ \text{odd}}}^{4-1} f(a+ih) + 2\sum_{\substack{i=2 \\ \text{even}}}^{4-2} f(a+ih) + f(b) \right]$$

$$= \frac{-47}{3}\left[f(80) + 4f(33) + 4f(-61) + 2f(-14) + f(-108) \right]$$

$$= 1.124733 \times 10^{-3}$$

$$\therefore \Delta D = -0.01391$$

b) Find the relative approximate error between the result of 2-segment and 4-segment Simpson's 1/3 rule. Analyze the reason for getting such low error. 4+3 (CO3) (PO2)

Error in Multiple-segment Simpson's 1/3 rule

The true error in a single application of Simpson's 1/3rd Rule is given[1] by

$$E_t = -\frac{(b-a)^5}{2880} f^{(4)}(\zeta), \quad a < \zeta < b$$

In multiple-segment Simpson's 1/3 rule, the error is the sum of the errors in each application of Simpson's 1/3 rule. The error in the n segments Simpson's 1/3rd Rule is given by

$$E_1 = -\frac{(x_2 - x_0)^5}{2880} f^{(4)}(\zeta_1), \quad x_0 < \zeta_1 < x_2$$

$$= -\frac{h^5}{90} f^{(4)}(\zeta_1)$$

$$E_2 = -\frac{(x_4 - x_2)^5}{2880} f^{(4)}(\zeta_2), \quad x_2 < \zeta_2 < x_4$$

$$= -\frac{h^5}{90} f^{(4)}(\zeta_2)$$

$$\vdots$$

$$E_i = -\frac{(x_{2i} - x_{2(i-1)})^5}{2880} f^{(4)}(\zeta_i), \quad x_{2(i-1)} < \zeta_i < x_{2i}$$

$$= -\frac{h^5}{90} f^{(4)}(\zeta_i)$$

$$\vdots$$

$$E_{\frac{n}{2}-1} = -\frac{(x_{n-2} - x_{n-4})^5}{2880} f^{(4)}\left(\zeta_{\frac{n}{2}-1}\right), \quad x_{n-4} < \zeta_{\frac{n}{2}-1} < x_{n-2}$$

$$= -\frac{h^5}{90} f^{(4)}\left(\zeta_{\frac{n}{2}-1}\right)$$

$$E_{\frac{n}{2}} = -\frac{(x_n - x_{n-2})^5}{2880} f^{(4)}\left(\zeta_{\frac{n}{2}}\right), x_{n-2} < \zeta_{\frac{n}{2}} < x_n$$

Hence, the total error in the multiple-segment Simpson's 1/3 rule is

$$= -\frac{h^5}{90} f^{(4)}\left(\zeta_{\frac{n}{2}}\right)$$

$$E_t = \sum_{i=1}^{\frac{n}{2}} E_i$$

$$= -\frac{h^5}{90} \sum_{i=1}^{\frac{n}{2}} f^{(4)}(\zeta_i)$$

$$= -\frac{(b-a)^5}{90n^5} \sum_{i=1}^{\frac{n}{2}} f^{(4)}(\zeta_i)$$

$$= -\frac{(b-a)^5}{180n^4} \frac{\sum_{i=1}^{\frac{n}{2}} f^{(4)}(\zeta_i)}{\frac{n}{2}},$$

The term $\frac{\sum_{i=1}^{\frac{n}{2}} f^{(4)}(\zeta_i)}{\frac{n}{2}}$ is an approximate average value of $f^{(4)}(x)$, $a < x < b$. Hence

$$E_t = -\frac{(b-a)^5}{180n^4} \bar{f}^{(4)}$$

where

$$\bar{f}^{(4)} = \frac{\sum_{i=1}^{\frac{n}{2}} f^{(4)}(\zeta_i)}{\frac{n}{2}}$$

2. a) To simplify a model for a diode, it is approximated by a forward bias model consisting of a DC voltage, V_d, and a resistor, R_d. Table 1 contains the voltage vs current data that is collected for a small signal: (CO2) (PO1) 12+4

Table 1: Voltage vs Current

V (volts)	I (amps)
0.6	0.01
0.7	0.05
0.8	0.2
0.9	0.7
1.0	2.0
1.1	4.0

Current can be linearly regressed from voltage data as $I = B_0 + B_1 V$, where B_0 and B_1 are the constants of regression. Once B_0 and B_1 are known, the voltage and resistance for the diode can be computed as $V_d = -\frac{B_0}{B_1}$ and $R_d = \frac{1}{B_1}$.

Find the voltage and resistance for the diode.

$I = B_0 + B_1 V$

$$B_1 = \frac{n\Sigma x_i y_i - \Sigma x_i \Sigma y_i}{n\Sigma x_i^2 - (\Sigma x_i)^2}$$

$= 7.5143$

$B_0 = \bar{y} - B_1 \bar{x}$

$= 1.16 - B_1\, 0.85$

$= -5.227155$

$\Sigma x_i y_i = 7.231$

$\Sigma x_i \Sigma y_i = 5.1 \times 6.96$

$= 35.496$

$n\Sigma x_i^2 = 6 \times 4.5$

$(\Sigma x_i)^2 = (5.1)^2$

Thus, $I = -5.227155 + 7.5143\, V$

$V_d = \frac{-B_0}{B_1} = 0.6956$ $\quad R_d = \frac{1}{B_1} = 6.133\,08$

b) Explain how relative error can provide more useful insight compared to true error when analyzing the results of a numerical method. 9 (CO3) (PO2)

The relative error can provide more meaningful insights when analyzing the results of a numerical method, as it takes into account the magnitude of the values being approximated.

You have two numerical methods, and you're using each to approximate a different value. Method 1 approximates a value around 1,000,000 with a true error of 100. Method 2 approximates a value around 10 with a true error of 1. In terms of true error, Method 1 seems to have a larger error.

However, the relative error tells a different story. The relative error for Method 1 is 100/1,000,000 = 0.0001, or 0.01%. The relative error for Method 2 is 1/10 = 0.1, or 10%. So, while the true error for Method 1 was larger in absolute terms, the relative error for Method 2 is actually larger. This suggests that Method 1 is more accurate when you consider the scale of the values being approximated.

Mathematically, we can represent true error (E_t) and relative error (E_r) as follows:

Let's say the true value is "T" and the approximated value is "A".

True Error (E_t): $E_t = |T - A|$

Relative Error (E_r): $E_r = |(T - A) / T|$

Given two numerical methods 1 and 2 with their corresponding approximations A1, A2 and true values T1, T2:

If $E_{t1} > E_{t2}$, it doesn't necessarily mean that method 1 is less accurate than method 2. But if $E_{r1} > E_{r2}$, it does mean that method 1 is less accurate than method 2, considering the scale of the values being approximated.

3. a) Derive the Trapezoidal rule of Integration from both Calculus and Method of Coefficients. Analyze the geometrical interpretation of these two derivations. (PO2) 10+5 (CO1) (PO2)

Calculus method :

$$\int_a^b f(x)\,dx = \int_a^b f_1(x)\,dx \quad ; f(x) = a_0 + a_1 x$$

$$= \int_a^b (a_0 + a_1 x)\,dx$$

$$= a_0(b-a) + a_1 \frac{b^2 - a^2}{2} \quad \text{——— (1)}$$

Now, $f(a) = a_0 + a_1 a$ ——— (2)

$f(b) = a_0 + a_1 b$ ——— (3)

Solving (2) & (3) $\Rightarrow$

$$\frac{f(a) - f(b)}{a - b} = a_1 \quad \text{and} \quad f(a) = a_0 - \frac{f(a) - f(b)}{a - b} a$$

$$\Rightarrow a_1 = \frac{f(b) - f(a)}{b - a} \quad \Rightarrow a_0 = f(a) + \frac{a f(a) - a f(b)}{b - a}$$

$$= \frac{b f(a) - a f(a) + a f(a) - a f(b)}{b - a}$$

$$= \frac{b f(a) - a f(b)}{b - a}$$

Thus, $\int_a^b f(x)\,dx = \frac{f(b) - f(a)}{b - a}(b - a) + \frac{b f(a) - a f(b)}{b - a} \times \frac{(b + a)(b - a)}{2}$

$$= (b - a)\left[\frac{f(a) + f(b)}{2}\right]$$

Co-efficient method

$$\int_a^b f(x) = c_1 f(a) + c_2 f(b)$$

Now, Right hand side is the exact integral of $\int_a^b 1\,dx$ and $\int_a^b x\,dx$

Now, $\int_a^b 1\,dx = b - a = c_1 + c_2$ — (1)

And $\int_a^b x\,dx = \frac{x^2}{2} \Rightarrow \frac{b^2 - a^2}{2} = c_1 a + c_2 b$ — (2)

Now, (2) - a × (1) $\Rightarrow$

$$\frac{b^2 - a^2}{2} - ab + a^2 = c_2 b - c_2 a$$

$$\Rightarrow \frac{b^2 - a^2 - 2ab + 2a^2}{2} = c_2 (b - a)$$

$$\Rightarrow \frac{b^2 + a^2 - 2ab}{2} = c_2 (b - a)$$

$$\Rightarrow c_2 = \frac{b - a}{2}$$

Now, (1) $\Rightarrow$

$$c_1 = b - a - \frac{b - a}{2}$$

$$\Rightarrow \frac{2b - 2a - b + a}{2} = \frac{b - a}{2}$$

Thus,

$$\int_a^b f(x) = \frac{b-a}{2} f(a) + \frac{b-a}{2} f(b)$$

Geometrical interpretation

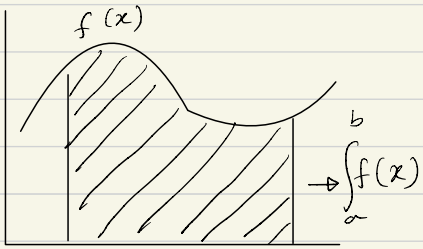


fig: Calculus method

Area = Integration

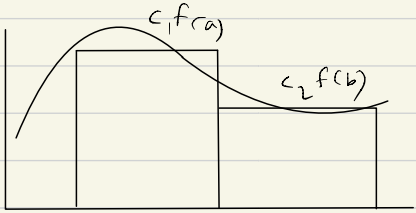


fig: Co-efficient method

Area = axb

b) Taking upto 5 terms of the Taylor series, find the value of $\cos(\pi)$ given that we know $\cos\left(\frac{\pi}{2}\right) = 0$. (PO2) 10 (CO2) (PO1)

Taylor series,

$$f(x+h) = f(x) + f'(x)h + \frac{f''(x)}{2!}h^2 + \frac{f'''(x)}{3!}h^3 + \frac{f''''(x)}{4!}h^4$$

Now, $x + h = \pi$

$\Rightarrow \frac{\pi}{2} + h = \pi$

$\Rightarrow h = \pi/2$

Now, $\cos(\pi/2) = 0$

$f'(x) = -\sin(\pi/2) = -1$

$f''(x) = -\cos(\pi/2) = 0$

$f'''(x) = \sin\left(\frac{\pi}{2}\right) = 1$

$f^4(x) = \cos\left(\frac{\pi}{2}\right) = 0$

$$\therefore \cos(\pi/2 + \pi/2) = 0 + (-1)\times\frac{\pi}{2} + 0 + \frac{1}{3!}\left(\frac{\pi}{2}\right)^3 + 0$$

$$= \frac{-\pi}{2} + \frac{\pi^3}{6\times 8}$$

$$= -0.92483$$

4. a) Thermistors are used to measure the temperature of bodies. These instruments are based on materials' change in resistance with temperature. To measure temperature, manufacturers provide you with a temperature vs. resistance calibration curve. If you measure resistance, you can find the temperature. Several observations with a thermistor are shown in Table 2. (PO1) 11 (CO2) (PO1)

Table 2: Resistance vs Temperature

R (ohm)	T (°C)
1101	25.113
911.3	30.131
636	40.12
451.1	50.128

Determine the temperature for 754.8 ohms using a 1st order Lagrange polynomial.

$$f(x) = \sum_{i=0}^{n} L_i f(x_i) \quad ; \quad L_i = \prod_{\substack{j=0 \\ j \neq i}}^{n} \frac{x - x_j}{x_i - x_j}$$

Now, $n = 1$; for 754.8, we choose (911.3, 30.131) and (636, 40.12)

$$L_0 = \prod_{\substack{j=1 \\ j \neq 0}}^{1} \frac{x - 636}{911.3 - 636}$$

$$L_1 = \prod_{\substack{j=1 \\ j \neq 1}}^{1} \frac{x - 911.3}{636 - 911.3}$$

Now, $f(x) = L_0 f(911.3) + L_1 f(636)$

$= L_0 \times 30.131 + L_1 \times 40.12$

$\therefore f(754.8) = 35.8095$

b) Using the previous data, determine the result for the same resistance (754.8 ohms) using a 2nd order Lagrange polynomial and find the absolute relative approximate error. 11+3 (CO3) (PO2)

Points are $(911.3, 30.131)$, $(636, 40.12)$, $(451.1, 50.128)$

$$L_0 = \frac{x - 636}{911.3 - 636} \times \frac{x - 451.1}{911.3 - 451.1}$$

$$L_1 = \frac{x - 911.3}{636 - 911.3} \times \frac{x - 451.1}{636 - 451.1}$$

$$L_2 = \frac{x - 911.3}{451.1 - 911.3} \times \frac{x - 636}{451.1 - 636}$$

Now, $f(x) = L_0 \times 30.131 + L_1 \times 40.12 + L_2 \times 50.128$

$\therefore f(754.8) = 35.0886$

$$\therefore \text{ARE} = \frac{|35.0886 - 35.8095|}{35.0886} \times 100\%$$

$$= 2.055\%$$

5. a) What is the minimum number of points needed to evaluate the integral for Simpson's 3/8 rule? Derive the Multiple Segment Simpson's 3/8 rule. (PO2) 3+10 (CO2) (PO1)

It requires minimum 4 points as it uses 3rd order

Now, $f(x_i) = a_0 + a_1 x_i + a_2 x_i^2 + a_3 x_i^3$

Using langrangian interpolation,

$$\int f(x) = L_0 f(x_0) + L_1 f(x_1) + L_2 f(x_2) + L_3 f(x_3)$$

$$\Rightarrow \frac{3h}{8}\left[f(x_0) + 3f(x_1) + 3f(x_2) + f(x_3) \right]$$

For multi segment,

$$h = \frac{b-a}{n}$$

Now, $I = \int_a^b f(x)\, dx$

$$= \int_{x_0=a}^{x_3} f(x)\, dx + \int_{x_3}^{x_6} f(x)\, dx + \cdots + \int_{x_{n-3}}^{x_n = b} f(x)\, dx$$

$$= \frac{3h}{8}\left[f(x_0) + 3f(x_1) + 3f(x_2) + f(x_3) \right]$$
$$+ \frac{3h}{8}\left[f(x_3) + 3f(x_4) + 3f(x_5) + f(x_6) \right]$$
$$+ \frac{3h}{8}\left[f(x_{n-3}) + 3f(x_{n-2}) + 3f(x_{n-1}) + f(x_n) \right]$$

$$= \frac{3h}{3}\left[f(x_0) + 3 \sum_{i=1,4,7}^{n-2} f(x_i) + 3 \sum_{i=2,5,8}^{n-1} f(x_i) + 2 \sum_{i=3,6,9}^{n-3} f(x_i) + f(x_n) \right]$$

b) Suppose you are given the following data: 3×4 (CO2) (PO1)

Table 3: x vs F(x)

x	0	0.5	1.0	1.5	2.0
F(x)	0	0.19	0.26	0.29	0.31

Find the Forward Difference, Backward Difference and Central Difference approximations of $F'(1)$ using $h = 0.5$.

x	$f(x)$	Forward $\frac{f(x_{i+1}) - f(x_i)}{h}$	Backward $\frac{f(x_i) - f(x_{i-1})}{h}$	Central $\frac{f(x_{i+1}) - f(x_{i-1})}{2h}$
0	0	0.38	—	—
0.5	0.19	0.14	0.38	0.26
1.0	0.26	0.06	0.14	0.10
1.5	0.29	0.04	0.06	0.05
2.0	0.31	—	0.04	—

6. Suppose you are given the following matrix A:

$$A = \begin{bmatrix} 3 & 1 & 2 \\ 1 & 2 & 3 \\ 2 & 3 & 1 \end{bmatrix}$$

a) Now, answer the following questions:

i) Analyze the properties of the matrix A to determine whether it can have complex eigenvalues. 4 (CO1)

Here, $|A_{11}| = 2\times1 - 3\times3 \rightarrow$ real

$|A_{12}| = 1\times1 - 3\times3 \rightarrow$ real

$|A_{13}| = 1\times3 - 2\times2 \rightarrow$ real

} Thus eigenvalues will also be real

ii) Find the eigenvalues and eigenvectors of the matrix A. 12

$$A = \begin{bmatrix} 3 & 2 & 1 \\ 1 & 2 & 3 \\ 2 & 3 & 1 \end{bmatrix} \quad ; \quad Av = \lambda v$$

$$\Rightarrow (A - \lambda I)v = 0$$

$$\Rightarrow A - \lambda I = 0$$

$$\Rightarrow \begin{bmatrix} 3-\lambda & 2 & 1 \\ 1 & 2-\lambda & 3 \\ 2 & 3 & 1-\lambda \end{bmatrix} = 0$$

$$\Rightarrow (3-\lambda)\{(2-\lambda)(1-\lambda) - 9\} - 2\{(1-\lambda) - 6\} + 1\{3 - 2(2-\lambda)\} = 0$$

$$\Rightarrow (3-\lambda)\{2 - 2\lambda - \lambda + \lambda^2 - 9\} - 2\{-\lambda - 5\} + 1\{-1 + 2\lambda\} = 0$$

$$\Rightarrow (3-\lambda)(\lambda^2 - 3\lambda - 7) + 2\lambda + 10 + 2\lambda - 1 = 0$$

$$\Rightarrow 3\lambda^2 - 9\lambda - 21 - \lambda^3 + 3\lambda^2 + 7\lambda + 2\lambda + 10 + 2\lambda - 1 = 0$$

$$\Rightarrow -\lambda^3 + 6\lambda^2 + 2\lambda - 12 = 0$$

$$\Rightarrow \lambda = 6, \pm\sqrt{2}$$

For, $\lambda = 6$

$$\begin{bmatrix} -3 & 2 & 1 \\ 1 & -4 & 3 \\ 2 & 3 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$\frac{x_1}{20-9} = \frac{-x_2}{-5-6} = \frac{x_3}{3+8}$$

$$\Rightarrow \frac{x_1}{11} = \frac{x_2}{11} = \frac{x_3}{11} \Rightarrow V_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} k_1$$

For $\lambda = \sqrt{2}$

$$\begin{bmatrix} 3-\sqrt{2} & 2 & 1 \\ 1 & 2-\sqrt{2} & 3 \\ 2 & 3 & 1-\sqrt{2} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$\frac{x_1}{-5-3\sqrt{2}} = \frac{-x_2}{-5-\sqrt{2}} = \frac{x_3}{-1+2\sqrt{2}} \Rightarrow V_2 = \begin{bmatrix} \frac{13\sqrt{2}-17}{7} \\ -\frac{11\sqrt{2}+9}{7} \\ 1 \end{bmatrix} k_2$$

Similarly, $V_3 = \begin{bmatrix} \frac{13\sqrt{2}-17}{7} \\ \frac{11\sqrt{2}-9}{7} \\ 1 \end{bmatrix} k_3$

b) Find the LU Decomposition of the matrix A.

$$A = \begin{bmatrix} 3 & 2 & 1 \\ 1 & 2 & 3 \\ 2 & 3 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

$$= \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ l_{21}u_{11} & l_{21}u_{12} + u_{22} & l_{21}u_{13} + u_{23} \\ l_{31}u_{11} & l_{31}u_{12} + l_{32}u_{22} & l_{31}u_{13} + l_{32}u_{23} + u_{33} \end{bmatrix}$$

Now,

$u_{11} = 3, \quad u_{12} = 2, \quad u_{13} = 1,$

$l_{21} = \frac{1}{3}, \quad u_{22} = 2 - \frac{1}{3} \times 2, \quad = \frac{4}{3}$

$u_{23} = 3 - \frac{1}{3} \times 1, \quad = \frac{8}{3}, \quad l_{31} = \frac{2}{3},$

$l_{32} = \frac{3 - \frac{2}{3} \times 2}{\frac{4}{3}} = 5/4$

$u_{33} = -3$

$$A = L\,U$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{3} & 1 & 0 \\ \frac{2}{3} & 5/4 & 1 \end{bmatrix} \begin{bmatrix} 3 & 2 & 1 \\ 0 & \frac{1}{3} & \frac{8}{3} \\ 0 & 0 & -3 \end{bmatrix}$$

Example 3

A ball at 1200K is allowed to cool down in air at an ambient temperature of 300K. Assuming heat is lost only due to radiation, the differential equation for the temperature of the ball is given by

$$\frac{d\theta}{dt} = -2.2067\times10^{-12}\left(\theta^4 - 81\times10^8\right),\ \theta(0) = 1200\text{K}$$

where θ is in K and t in seconds. Find the temperature at $t = 480$ seconds using Euler's method. Assume a step size of $h = 240$ seconds.

Here, $f(\theta, t) = -2.2067\times10^{-12}\left(\theta^4 - 81\times10^8\right)$

t	θ	$\theta_{i+1} = \theta_i + f(\theta_i, t_i)h$
0	1200	$= 1200 + f(1200, 0)\times240$ $= 106.09\ k$
$0+240 = 240$	106.09	$= 106.09 + f(106.09, 240)\times240$ $= 110.32\ k$
$240+240 = 480$	110.32	———

Example 4

Find an approximate value of

$$\int_5^8 6x^3 dx$$

using Euler's method of solving an ordinary differential equation. Use a step size of $h = 1.5$.

$f(x, y) = 6x^3$

x_i	y_i	$y_{i+1} = y_i + f(x_i, y_i)h$
5	0	$0 + (6 \times 5^3) \times 1.5 = 1125$
$5 + 1.5 = 6.5$	1125	$1125 + (6 \times 6.5^3) \times 1.5 = 3596.625$
$6.5 + 1.5 = 8$	3596.625	———

Example 3

A ball at 1200 K is allowed to cool down in air at an ambient temperature of 300 K. Assuming heat is lost only due to radiation, the differential equation for the temperature of the ball is given by

$$\frac{d\theta}{dt} = -2.2067\times10^{-12}\,(\theta^4 - 81\times10^8)$$

where θ is in K and t in seconds. Find the temperature at $t = 480$ seconds using Runge-Kutta 2nd order method. Assume a step size of $h = 240$ seconds.

Heun $\Rightarrow$ $a_2 = \frac{1}{2}$, $a_1 = \frac{1}{2}$, $p = 1$, $q = 1$

$k_1 = f(x_i, y_i) = -2.2067\times10^{-12}\,(\theta^4 - 81\times10^8)$

$k_2 = f(x_i + ph,\ y_i + qk_1h) = f(x_i + h,\ y_i + k_1h)$

$\theta_{i+1} = \theta_i + (a_1k_1 + a_2k_2)h$

$= \theta_i + (\frac{1}{2}k_1 + \frac{1}{2}k_2)h$

t	θ	k_1	k_2	$\theta_{i+1} = \theta_i + (\frac{1}{2}k_1 + \frac{1}{2}k_2)h$
0	1200	−4.558	0.0176	$1200 + (\frac{1}{2}k_1 + \frac{1}{2}k_2)\times240 = 655.152$
$0+240=240$	655.152	−0.389	−0.202	1129.08
$240+240=480$	1129.08	—	—	—

Example 2

Given

$$\frac{d^2y}{dt^2}+2\frac{dy}{dt}+y=e^{-t},\ y(0)=1,\ \frac{dy}{dt}(0)=2,$$ find by Euler's method

a) $y(0.75)$

b) the absolute relative true error for part(a), if $y(0.75)\big|_{exact}=1.668$

c) $\frac{dy}{dt}(0.75)$

Use a step size of $h=0.25$.

a) $\frac{d^2y}{dt^2}+2\frac{dy}{dt}+y=e^{-t}$

$\Rightarrow \frac{dz}{dt}+2z+y=e^{-t}$

$\Rightarrow \frac{dz}{dt}=e^{-t}-2z-y$

$\Rightarrow f(t_i, y_i, z_i)=e^{-t}-2z_i-y_i$

t	y_i	z_i	$y_{i+1} = y_i + f_1(t, y, z)h$	$z_{i+1} = z_i + f_2(t, y_i, z_i)h$
0	1	2	$1 + \frac{dy}{dx} \to f(t,y,z)$ $1 + z_i h = 1 + 2 \times 0.25$ $= 1.5$	$2 + (1 - 2 \times 2 - 1) \times 0.25$ $= 1$
$0 + 0.25$ $= 0.25$	1.5	1	$1.5 + 1 \times 0.25 = 1.75$	0.3197
$0.25 + 0.25$ $= 0.50$	1.25	0.3197	$1.75 + 0.3197 \times 0.25$ $= 1.8299$	$0.31972 + (-0.4457)$ $= -0.12$
$0.5 + 0.25$ $= 0.75$	1.8299	-0.126	———	———

b) $$ALE = \left| \frac{True - Abs}{True} \right| \times 100\%$$

$$= \left| \frac{1.668 - 1.830}{1.668} \right| \times 100\%$$

$$= 9.71\%$$

c) $$\frac{dy}{dt}(0.75) = z_3 = 1.830$$

Example 3

Given

$$\frac{d^2y}{dt^2}+2\frac{dy}{dt}+y=e^{-t}, y(0)=1, \frac{dy}{dt}(0)=2,$$

find by Heun's method

a) $y(0.75)$

b) $\frac{dy}{dx}(0.75)$.

Use a step size of $h=0.25$.

$$\frac{d^2y}{dt^2}+2\frac{dy}{dt}+y=e^{-t}$$

$$\Rightarrow \frac{dz}{dt}+2z+y=e^{-t}$$

$$\Rightarrow \frac{dz}{dt}=e^{-t}-2z-y$$

$$\therefore f(t_i, y_i, z_i)=e^{-t}-2z_i-y_i$$

$$y_{i+1}=y_i+\left(\frac{1}{2}k_1^y+\frac{1}{2}k_2^y\right)h$$

$$k_1^y=f_1(t, y_i, z_i)=z_i$$

$$k_2^y=f_1(t+h, y_i+hk_1^y, z_i+hk_1^z)$$

$$z_{i+1}=z_i+\left(\frac{1}{2}k_1^z+\frac{1}{2}k_2^z\right)h$$

$$k_1^z=f_2(t, y_i, z_i)=e^{-t}-2z-y$$

$$k_2^z=f_2(t+h, y_i+hk_1^y, z_i+hk_1^z)$$

t	y_i	z_i	k_1^y	k_1^z	k_2^y	k_2^z	y_{i+1}	z_{i+1}
0	1	2	2	-4	1	-2.7212	1.375	1.160
0.25	1.375	1.460						

b) $\frac{dy}{dx}(0.75) = z_3$